# Code–Monomial Matrix Groups

Generalized Molien formula, dual-code proof, and explicit verification

Computational verification note

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**Outcome.** The proposed weighted-cycle formula and the dual-code orbit formula are correct under the stated prime-alphabet convention. Explicit matrices for three groups of orders 8, 24, and 18 were used to check all 57 coefficients through total degree five. All agree. Three independent determinant evaluations agree to machine precision (maximum error 0).

## 1 Master coefficient identity

Let $G \le GL(V)$ be finite and put

$$\mathfrak{M}_{G,V}(\mathbf{t}) = \frac{1}{|G|}\sum_{g\in G}\prod_{j\ge 1}\det(I - t_j g)^{-1}.$$

If $g$ has eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\det(I - tg)^{-1} = \prod_i (1 - t\lambda_i)^{-1} = \sum_{d\ge 0}\chi_{\mathrm{Sym}^d V}(g)t^d.$$

It follows that, for $\tau = (\tau_1, \dots, \tau_\ell)$,

$$[t_1^{\tau_1}\cdots t_\ell^{\tau_\ell}]\mathfrak{M}_{G,V} = \dim\left(\bigotimes_{s=1}^{\ell}\mathrm{Sym}^{\tau_s} V\right)^G. \tag{1}$$

Indeed the coefficient is the trace of the Reynolds projector $|G|^{-1}\sum_g g$ on the displayed tensor product. Under the usual symmetric-function packaging this is the coefficient of $m_\tau$.

## 2 Code–monomial theorem

Let $r$ be prime, $C \le \mathbf{F}_r^n$, and let

$$H \le \mathrm{Aut}_{\mathrm{perm}}(C) := \{\pi \in S_n : \pi(C) = C\}$$

be a coordinate-permutation automorphism group. Put $\zeta = e^{2\pi i/r}$ and define

$$D_c = \mathrm{diag}(\zeta^{c_1}, \dots, \zeta^{c_n}), \qquad G_C = D_C \rtimes H.$$

Use $P_\pi e_i = e_{\pi(i)}$. On a coordinate cycle $O$ of $\pi$, the weighted shift $D_c P_\pi$ returns after $|O|$ steps with scalar $\zeta^{c(O)}$, where $c(O) = \sum_{i\in O} c_i$. Therefore

$$\det(I - tD_cP_\pi) = \prod_{O\in\mathrm{Cyc}(\pi)}(1 - \zeta^{c(O)}t^{|O|}). \tag{2}$$

Substitution into the master average proves

$$\boxed{\mathfrak{M}_{G_C,V}(\mathbf{t}) = \frac{1}{|C||H|}\sum_{\pi\in H}\sum_{c\in C}\prod_{j,O}(1-\zeta^{c(O)}t_j^{|O|})^{-1}.} \tag{3}$$

### 2.1 Why the coefficient is an orbit count

A monomial basis of $\bigotimes_s \operatorname{Sym}^{\tau_s} V$ is indexed by arrays $A=(a_{si})\geq 0$ with $\sum_i a_{si}=\tau_s$. Set

$$w(A)_i = \sum_s a_{si} \pmod r.$$

The diagonal matrix $D_c$ multiplies the basis vector by $\zeta^{c\cdot w(A)}$. Character orthogonality says it survives all diagonal matrices precisely when $w(A)\in C^\perp$. The remaining group $H$ permutes columns, so invariants in this permutation module are constant on $H$-orbits. Hence

$$\boxed{[m_\tau]\mathfrak{M}_{G_C,V} = \#\big(H\backslash\{A:\sum_i a_{si}=\tau_s,\ w(A)\in C^\perp\}\big).} \tag{4}$$

This also proves integrality without relying on cancellation in (3).

The proof uses only coordinate permutations and therefore does not cover the full monomial or semilinear automorphism group of a code. If coordinate scalings or field automorphisms are included, their characters must be incorporated into the diagonal filter and the column-orbit argument must be replaced by the corresponding semilinear action. The verifier rejects an automorphism entry that is not a coordinate permutation.

## 3 Independent verification method

The implementation uses routes that share the group input but not the coefficient algorithm:

1. Enumerate every complex matrix $D_cP_\pi$. Obtain $\chi_{\operatorname{Sym}^d V}(g)$ from the eigenvalue series $\prod_i(1-t\lambda_i)^{-1}$ and average products of characters.
2. Enumerate exponent arrays, impose $c\cdot w(A)=0$ for every $c\in C$, and count column-permutation orbits. No eigenvalues occur in this route.
3. At $(t_1,t_2)=(0.07,0.11)$ compare the direct matrix determinant average with (3), evaluated solely from weighted coordinate cycles.

## 4 Representative cases and results

Every partition of every total degree 0 through 5 was checked (19 coefficients per group).

| Case | $\dim V$ | $\lvert C\rvert$ | $\lvert H\rvert$ | checks | result |
|---|---|---|---|---|---|
| $W(B_2)$, full binary code | 2 | 4 | 2 | 19 | **PASS** |
| $W(D_3)$, even binary code | 3 | 4 | 6 | 19 | **PASS** |
| ternary repetition code $\rtimes S_3$ | 3 | 3 | 6 | 19 | **PASS** |

| Case | (2) | (3) | $(2,1)$ | $(1,1,1)$ |
|---|---|---|---|---|
| $W(B_2)$ | 1 | 0 | 0 | 0 |
| $W(D_3)$ | 1 | 1 | 1 | 1 |
| ternary repetition | 0 | 3 | 4 | 5 |

The numerical generalized-Molien values were respectively 1.025494342880870, 1.028638382342067, and 1.010621973731615 by both routes.

## 5 Scope and reusable tool

For $\mathbf{F}_{p^f}$ with $f > 1$, the notation $c \mapsto \zeta^c$ is not a faithful additive embedding of the whole additive group into one cyclic root group. Formula (4) remains valid after replacing $C^\perp$ by the annihilator of the chosen diagonal-character group. The reusable computational idea is to test (3) numerically while testing (4) exactly: agreement localizes errors to either the weighted-cycle determinant or the character-orthogonality filter.